FCS Innovation Academy

Course Catalog 2026-2027

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Who are we?

In addition to the core curriculum, FCS Innovation Academy is a STEM magnet high school that provides an educational experience that will help push students to be creators and problem-solvers for a continuously changing world.

Philosophy

Innovation Academy's philosophy emphasizes student collaboration, relevant learning, rigorous courses, and flexible scheduling to allow for deeper and more personalized learning experiences. Students will experience the curriculum through a project-based learning environment where teachers and students work together to solve practical and complex issues. Our goal is for students to receive a rich and relevant educational experience that prepares them to be empathetic leaders in our ever-changing society.

IA Student Traits

Innovation Academy is a different school, and we are looking for students who:

- Get the itch to figure something out that they don't understand.
- Like to get their hands dirty, literally or figuratively.
- Like to take risks.
- Are problem-seekers and problem-solvers.
- Are willing to take chances, make mistakes, and learn from those mistakes.
- Are humble enough to ask questions and seek answers from others, but confident enough to keep going when things aren't easy.
- Work well together and can lead a group and be led by others.
- Are nervous about trying something new but are excited by the opportunities it will give them.
- Are ok with the first draft not being the last draft and continuing to iterate.

As our students engage with our curriculum, work through their career pathways, and solve difficult and complex problems with what they have learned in class, they will gain the skills they need to be successful in the world after Innovation Academy.

The IA Graduate will:

- Be comfortable taking risks.
- Have the confidence to tackle difficult problems and the humility to seek answers from others when necessary.
- Have the ability to create innovative solutions to complex problems.
- Think critically and analyze information.
- See things from multiple perspectives and understand them in different contexts.
- Look at the world with curiosity and wonder.
- Work well in teams, leading when appropriate and following when necessary.
- Be kind to all others in our world.

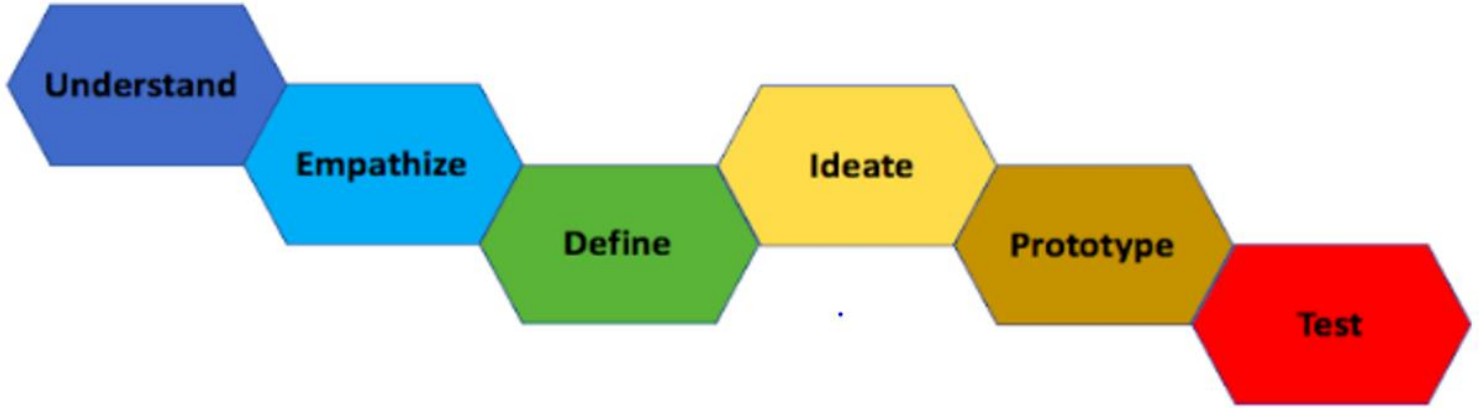
The IA Experience

In addition to a traditional high school curriculum, students will have the opportunity to expand their knowledge and experiences through rigorous applications of their learning:

- Problem solving using Design Thinking
- Choosing a Career Pathway
- Partnering with Georgia Tech's CEISMC program
- Completing two magnet requirements through the GT Block **and** the Pinnacle Project through the junior year research class

Design Thinking

Design Thinking is a method that uses a cyclical process to solve problems with the user in mind. Design Thinking uses a series of steps that creates specific work in each phase. The graphic below shows the various steps and how students would engage in each area. Students will use design thinking while participating in the AP Capstone project.



Understand:

Students are introduced to a challenge and work to research and understand the nuances of the challenge.

Empathize:

Students define the audience impacted by the challenge. Students conduct empathy interviews to understand the needs of the user and define the problem.

Define:

Students will frame research into an actionable problem statement to create a statement that will drive the process.

Ideate:

The ideate phase allows students to brainstorm ideas and cluster them into design concepts. No idea is a bad idea!

Prototype:

Students create low fidelity prototypes to present to their user audience as first draft of a solution to the problem.

Test:

Students test prototypes with the user audience. After gaining feedback about the prototype, students revert back to gain more empathy to refine and iterate on the prototype.

Career Pathways at IA

Each student will complete a pathway in one of 3 CTAE areas, Healthcare Science, Engineering, or Information Technology. These CTAE pathways will provide relevance to student learning and incorporate a hands-on implementation of projects for standards from the core academic areas.

Health Science

Health Science offers a variety of health care related skills and concentrations, all of which include a state-of-the-art simulation center where we can LEARN BY DOING. Our students can gain knowledge and skills in a safe, non-threatening, experiential environment, giving them an opportunity to learn through hands-on training. Students will learn to work as a team, learn effective communication skills, and develop decision making and problem-solving skills. Students will work in a more realistic learning environment and immerse themselves in learning by being actively engaged. They will be able to see the result of their decisions and actions and the consequences that occur.

Using this technology, we will build on the traditional educational framework while incorporating new methods. Simulation is an alternate and unique way to enhance teaching and learning. Simulation is a way for us to replicate what might occur in real life. For example, students will not just learn about lung sounds; they can actually get to listen to them on these simulators or mannequins. It will help students understand healthcare in a technologically evolving and demanding time. What an amazing and fun way to learn about what you love!

The way the classes and labs are set up will allow us to work with students independently and in different size groups. Through simulation we will work with our business partners to develop areas that should be the highest priority for students to experience.

Engineering

In engineering, we will use CAD, code, and calculations to solve problems. We will empower students with a foundation in CAD and 3D printings, electrical and pneumatic systems, and culminate with programming in C++. The most rewarding part is using these skills to create functional autonomous prototypes to solve the problem provided by a business partner. Students will utilize our state-of-the-art facility to ideate and prototype as they progress through the design process. As students start with raw materials, they will learn to operate the tools and machinery needed to create custom parts with the correct tolerances for their prototypes. From plastics and fabrics, to lumber or metal, “measure twice and cut once” will often be heard as students fabricate custom designs. These constraints will guide them in making tough decisions requiring them to analyze tradeoffs and make compromises that are common in real world engineering.

Informational Technology

The computer science industry thrives on analytical, critical problem-solvers. Computer engineers and programmers need these skills to succeed in their careers, but the best also excel at creative thinking and teamwork. At IA, students interested in computer science should focus on these core skills rather than whether they can already code. Many students shy away from programming because they don’t understand the language, yet they start new world languages without the same fear. If you’re good at teamwork, critical thinking, and willing to see things from different perspectives, you’ll likely excel in computer science.

Partnering with Georgia Tech's CEISMC program

The Center for Education Integrating Science, Mathematics and Computing (CEISMC) is a unit within the Office of the Provost at Georgia Tech. CEISMC serves as the primary connection point between Georgia Tech faculty and students and the preK-12 STEM education community.

CEISMC plays a key role in achieving the vision set forth in the Commission on Creating the Next (CNE) report by reducing the barriers between prek-12 and high education by offering intensive, innovative professional development to practicing teachers, creating highly engaging STEM in-school and extracurricular STEM experiences for K-12 students.

Fulton County Schools and Innovation Academy have partnered with CEISMC to create a dynamic and rigorous set of challenges and projects that are designed to incorporate core standards and relevance to learning. Students will use design thinking to solve these complex problems.

IA Magnet Requirement: GT Block

The GT Block course is based on a cross curricular, project-based framework designed by the CEISMC group at Georgia Tech specifically for Innovation Academy. This unique course combines the three CTAE departments within Innovation Academy: Information Technology, Engineering and Healthcare.

The GT Block covers the three introductory courses from all three departments. The block course is a two-class period block that meets each school day. Students will have a teacher from each department and will rotate through the three introductory courses within each unit. The GT Block course is unique to Innovation Academy and will allow students to experience true cross-curricular, design thinking projects while simultaneously earning course credit for all three introductory courses.

Introduction to Software Technology

This course is the foundational course for the Computer Science, Cybersecurity, Embedded Computing, and Game Design Pathways. This course is designed for high school students to understand, communicate, and adapt to a digital world as it impacts their personal life, society, and the business world. Exposure to foundational knowledge in hardware, software, programming, and web design are all taught with hands-on activities and project focused tasks. Students will not only understand the concepts but apply their knowledge to situations and defend their actions/decisions/choices through the knowledge and skills acquired in this course.

Introduction to Healthcare Science

Introduction to Healthcare Science is the foundational course and the prerequisite for all Health Science pathways. Pathways will culminate with the opportunity to sit for a certification and or licensing exam in the designated allied health field. Students will participate in a block course rotation that will consume two of their class periods. We will focus on critical thinking, ethical reasoning, effective communication, and self-directed learning. During the rotations, students will receive exposure to all three areas of study: Information Technology, Engineering, and Healthcare Science. Within the healthcare science rotation, students receive initial exposure to the many Allied Health Careers. Topics to be covered include employability skills, communication,

lifespan, legal and ethical considerations, wellness and clinical competencies, which are the skills specific to the pathway. The knowledge and skills obtained in the Introduction courses will provide a basis for advancement into subsequent CTAE classes. Our focus is to provide students with the competitive edge that will make them top candidates for either entry into the healthcare global marketplace and/or the post-secondary institution of their choice to continue their education and training.

Foundations of Engineering

Engineers solve problems to make our world a better place. Our students will experience this through the engineering design process. They will engage in problem solving in three different spheres: locally (school), regionally (community) and globally. Students will utilize our state-of-the-art facility to ideate and prototype as they progress through the design process. Moving from low fidelity cardboard prototypes, they will then transition to wood and 3D modeling and finally to designs of metal or 3D prints depending on the application. The facilities at IA allow students to experience realistic learning environments from safely operating wood working equipment to welding metal to 3D printing. Students will tangibly see the results of their work and experience real world constraints. These constraints will guide them in making tough decisions requiring them to analyze tradeoffs and make compromises that are common in real world engineering.

IA Magnet Requirement: Pinnacle Project

Both an opportunity and a requirement, all IA students complete a Pinnacle Project during their junior year. Designed as an experience that will offer offers students the opportunity to learn a topic broadly and deeply, then to complete students the opportunity to learn a topic broadly and deeply and then to complete full-cycle design thinking or scientific research that answers a question of their own design, the IA Pinnacle Project truly sets our students apart. To support the background research and active project work, IA offers two choices for this year-long course. You will see the Pinnacle Project class coded as Honors Research or AP Research.

Pinnacle Project: Honors Research

Honors Research is the premier capstone offering high rigor of process, added authenticity, and opportunity for real impact of the product. As an honors-level course, students experience research through the lenses of both the scientific method and design thinking. Over the course of the year, students will frame, execute and communicate their independent research work through IA's Pinnacle Project using industry best practices. Honors Research is analogous to AP Research in that research in any area - pathway, ELA, science, math, social sciences, arts - is supported. However, in AP Research, College Board does not allow partner projects or small group work to take on real-world problems collaboratively.

For the 25-26 School Year, applications-based special sections of Research are possible based on their Research Proposal submitted during the second semester of their Sophomore Pinnacle Prep Sessions. These focus on Science Experiments, the Arts, and other sections based on interest, capacity, and capability.

Pinnacle Project: AP Research

At IA, AP Research is offered primarily as a senior elective for students who wish to pursue further research and to complete the AP Capstone program. AP Research, the second course in the AP Capstone experience, allows students to deeply explore an academic topic, problem, issue, or idea of individual interest. Students design, plan, and implement a yearlong investigation to address a research question. Through this inquiry, they further the skills they acquired in the AP Seminar course by learning research methodology, employing ethical research practices, and accessing, analyzing, and synthesizing information. Students reflect on their skill development, document their processes, and curate the artifacts of their scholarly work through a process and reflection portfolio. The course culminates in an academic paper of 4,000–5,000 words, accompanied by a performance, exhibit, or product, and a presentation with an oral defense. The AP Capstone Diploma recognition is widely recognized by universities across Georgia and the U.S. for its rigor and is accepted by the University System of Georgia as an elective only.

IA Magnet Requirement: Pathway Completion

All students who graduate from Innovation Academy must complete at a minimum three credits (equal to three courses) in one of the pathways and complete the EOPA exam associated with that pathway. See the course progressions below for the three pathways.

Additional IA Magnet Requirements

All students at FCS Innovation Academy are considered COSA students by Fulton County Schools, Change of School Assignment. FCS board policy outlines requirements that students must meet to continue in COSA. All of our magnet requirements with further details can be found on our website: https://fcsinnovationacademy.fultonschools.org/counseling-academics/magnet-school-requirements

Dual Enrollment

The Dual Enrollment Program provides students the opportunity to earn high school and college credit at a participating eligible postsecondary institution in Georgia. Students may enroll on a part-time or full-time basis and take college courses at their high school, online, or on a post-secondary campus. The program provides standard tuition, mandatory fees, and required books to participating students. Students must complete the required Fulton County Schools dual enrollment paperwork by the March 31 deadline in order to participate in Summer or Fall dual enrollment classes. Any student wanting to participate in the Spring semester must complete paperwork no later than November 1 each year. For more information, please visit the IA Counseling website and/or www.gafutures.org. The program is offered during all terms of the school year.

Dual Enrollment Grade Level Eligibility:

11th & 12th Graders

- Eligible students may take any approved Dual Enrollment courses listed on the Ga Futures Course Directory, at an eligible participating postsecondary institution (USG, TCSG or private).

10th Graders

- Eligible students may enroll in approved Career, Technical and Agricultural Education (CTAE) courses listed on the Course Directory at a participating TCSG institution only.
- Eligible students who have a minimum SAT score of 1200 or minimum ACT composite score of 26 in a single national test administration, may enroll in any approved courses listed on the Course Directory at a TCSG, USG or private eligible participating postsecondary institution.

9th Graders

- Students in the 9th grade are not eligible to participate in the Dual Enrollment funding Program

Georgia Graduation Requirements

Students must earn 23 credits to graduate from high school:

Area of Study	Units Required
English/Language Arts	4
Mathematics	4
Science	4
Social Studies	3
CTAE and/or World Language/Latin/Fine Arts	3
Health/Physical Education	1
Electives	4
Total	23

*See Appendix 1 for course progressions for each content area.

**See Appendix 2 for the Counseling Team Graduation Status Report that is used to track courses and credits over the four years of high school.

How Courses Are Chosen and How Changes Are Made

There are many steps and considerations when choosing courses for a student to take for the upcoming school year. Students in partnership with their counselor should consider graduation requirements, workload management, interests, and pathway focus. We offer several opportunities during the process for teachers, students, and parents to have input on the course selection process.

Step 1 (January): Teachers speak with students and make recommendations on the next course for the students in their content area. Teachers use assessment mastery, work ethic, and knowledge of the next course to make these recommendations.

Step 2 (February): Students and parents look at the recommendations for the next year and make any changes needed based on student workload management, interest, and need. Changes include leveling up (see note about waivers below) and leveling down from the teacher recommendations. Students and parents follow the course request change process put in place so counselors are able to make the requested changes.

Step 3 (March): Students and parents take a SECOND look at the course list for the student and make any changes needed based on student workload management, interest, and need. Changes include leveling up (see note about waivers below) and leveling down from the teacher recommendations. Final course loads are sent to students and parents in May before leaving for the spring.

***Note:** There are no course changes over the summer unless a FVS summer course is taken, or a Dual Enrollment course is taken.

Step 4 (First 5 days of school): This is the final period of time that a student or parent can initiate a course change following the counseling team protocol. This does not include emailing or calling the counseling office; the submission is made electronically. Due to the triple verification in the spring, there should be limited changes at this time. NO SCHEDULE CHANGES ARE MADE FOR TEACHER PREFERENCE OR CLASS TIMING PREFERENCE.

Step 5 (Week 2, 3, and 4 of the school year): Teachers may reach out to parents and students about possible needs for a student to move courses. This decision is made on a case-by-case basis using student data and is only initiated by the teacher, not the parent or student. The parent and student may agree or not agree with the change. <u>All changes may not be possible due lack of open seats in classes.</u>

Fall to Spring Semester Course Changes: These types of course changes are <u>initiated by the teacher</u> based on grades, skills, and knowledge seen over the course of the fall semester. These changes are also very limited due to space in classes. Teachers will initiate a call or email with parents if a change is warranted.

If a student chooses to move into a higher level course than what they are recommended for, a course waiver agreement will need to be signed and kept on file. Course waiver agreements lock a student into the course for the year.

If a student chooses to enroll in the following combinations of classes, we consider this a highly rigorous courseload:

- 3 or more AP courses
- Combination of 4 or more Honors and AP courses,
- 3 Dual Enrollment courses
- Combination of 3 or more Honors, AP, and Dual Enrollment courses

We ask that parents and students are very careful when overloading a schedule with too many rigorous classes and make smart decisions for the well-being of the student.

Due to the extensive nature of the courses here at Innovation Academy, we recommend that students do not take more than one virtual course through FVS and/or GAVS. Our data has shown that the independent nature of online courses combined with our in person offerings require a high level of investment and accountability.

Students and parents will receive multiple opportunities to explore course options and course workload amounts to make the best decision for their individual student. Other students or families should not influence course load decisions.

IA Course Catalog

IA Unique Dual Courses Based on Interest

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics
Element Six: An Interdisciplinary Exploration in Medicine, Arts & Meaning	Multicultural Literature: 23.0670001 23.0670002 Patient Care Fundamentals: 25.4360001 25.4360002	Y	11th + 12th	Lit 1 & 2 credit – 90 or above Teacher Recommendation Healthcare Pathway: Completion of Intro to Healthcare & Essentials of Healthcare	**Scope**: contemporary works of narrative art from patient & provider perspectives coupled with unique clinical skill sets of patient care **Focal points:** a) Exploring and building the wider intersections of literature and medicine b) Exploring and building an understanding of range of expressions from provider and patient perspectives in art c) Reading and analyzing memoir, short stories, and poetry from these dual perspectives d) Reading and analyzing secondary-source nonfiction, fiction, poetry, drama, visual art, music, and film/television reflective of these perspectives e) Contribute original pieces of literature in workshop setting f) Conduct field interviews with medical professionals g) Participation in High-fidelity simulation lab training that enhances understanding of patient care, interpersonal skills, and systems-based practice h) Demonstration and application of a broad range of medical skills including phlebotomy, electrocardiograms, vital signs, and wound care g) Integrating clinical judgement and interprofessional perspectives related to common disease processes across the lifespan

CTAE: Engineering

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics

Foundations of Engineering & Technology *Embedded in the GT Block class	21.4250001 21.4250002	Y	9	None	How to think like an engineer, use a decision matrix, Gantt Charts, collect and present data in excel, use computer aided design to create virtual models, 3D printing of models, DC circuits, introduction to robotics, programming in C, measuring and dimensioning techniques, tool safety, heat transfer, physical property analysis, free body diagrams, and presentation skills
Engineering Concepts	21.4710001 21.4710002	Y	10-12	Foundations of Engineering & Technology	Advanced CAD, CAD assemblies, mechanical mates and animation, complex planes, tolerances, datum dimensioning, calculations of DC circuits, waves, soldering, statistical analysis of data, advanced safety training for use of power tools to work with lumber, use of laser engravers, programming in C++, robotics, mechanical advantage, and simple machines, OSHA certification
Engineering Applications	21.4720001 21.4720002	Y	11	Foundations of Engineering & Technology AND Engineering Concepts	Creating advanced views in CAD, full GD&T, rendering, parametric modeling, pneumatic systems, robotics, computer numeric control, advanced safety training for fabrication with nonferrous metals - primarily aluminum, milling, plasma cutting, TIG welding, compound machines, programming in C++, preparation for CAD certification exam in SolidWorks, preparation for engineering pathway exam
Research, Design, and Project Management	21.4610001 21.4610002	Y	12	Engineering Applications	Advanced architectural CAD software Revit, advanced models and rendering, vr compatibility, beam deflection calculations, truss analysis, heat transfer, surveying, site preparation, water runoff calculations, preparation for CAD certification exam in Revit, research and application of advanced communication methods

The Engineering department has started a cohort of students who are take Engineering Concepts and Engineering Applications with a heavy emphasis in Mechatronics. This course provides an in-depth exploration of mechatronics, the intersection of mechanical, electrical, and computer systems. Students will gain hands-on experience in designing and building automated systems by mastering concepts such as circuits and electrical components, pneumatics, CAD modeling, fabrication techniques, and soft robotics. The course emphasizes critical thinking, creativity, and problem-solving through real-world applications of mechatronics in automation and robotics. By the end of the course, students will be equipped to prototype and control advanced systems, preparing them for careers in engineering and technology fields. Students apply as 9th graders to work in this cohort as sophomores and juniors.

CTAE: Information Technology

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics
Introduction to Software Technology *Embedded in GT Block*	11.4460001 11.4460002	Y	9	None	Introduction to Software Technology is the foundational course for Artificial Intelligence, Computer Science, Cybersecurity, Embedded Computing, and Game Design Pathways. This course is designed for high school students to understand, communicate, and adapt to a digital world as it impacts their personal life, society, and the business world. Exposure to foundational knowledge in programming languages, software development, app creation, and user interfacing applications are all taught with hands-on activities and project-focused tasks.
AP Computer Science Principles	11.0190001 11.0190002	Y	10-11	Intro to Software Technology in the GT Block	AP Computer Science Principles is equivalent to a first-semester college computing course. Students build computational thinking, including data analysis and algorithmic skills, while focusing on creativity. They learn to develop computational artifacts, explore personal interests, and enhance communication and collaboration skills. Through individual and group work, students discuss and write about the significance of these problems and their impact on society and the world. This is a 2nd year pathway course for the Computer Science, Game Desing, and Internet of Things Pathways
Foundations of Artificial Intelligence	11.4430001 11.4430002	Y	10	GT Block IT Teacher Recommendation AND Advanced Algebra Honors – unweighted 88 or above OR Geometry Honors – unweighted 88 or above	Foundations of Artificial Intelligence is the introductory course to the Artificial Intelligence pathway, which prepares students to better understand common Artificial Intelligence applications and apply their knowledge to solve real-world problems using advanced technologies. This introductory course explores the foundations of Artificial Intelligence in society and the workplace, including programming, data science, mathematical reasoning, creative problem solving, ethical reasoning, and real-world applications of Artificial Intelligence. Students will learn the foundational skills to understand how to both interact and develop Artificial Intelligence solutions in a variety of settings. Corequisites: AP Computer Science Principles This is the 10th grade course for the Artificial Intelligence Pathway
Artificial Intelligence Concepts	11.4440001 11.4440002	Y	11	Foundations of Artificial Intelligence	Artificial Intelligence Concepts is the second course in the Artificial Intelligence pathway. Students will build upon their foundational understanding of Artificial Intelligence from course one and deepen their understanding of Artificial Intelligence and machine learning concepts.

					The course continues to explore the elements from course one, with an emphasis on using data science and programming to solve problems. The course requires students to explore real-world applications, potential benefits, and problems, as well as ethical considerations. Students will work to design potential solutions to problems using learning from this course. This is the 11th grade course for the Artificial Intelligence Pathway
Artificial Intelligence Applications	11.4450001 11.4450002	Y	12	Artificial Intelligence Concepts	Artificial Intelligence Applications is the third course in the Artificial Intelligence pathway. This course deepens students' understanding from courses one and two by requiring students to design and apply Artificial Intelligence solutions. The course focuses on creating functional, real-world applications of Artificial Intelligence using machine learning, neural networks, or other models. Students will apply professional software development tools and processes to solve problems using elements of machine learning in software and embedded computing. This is the 12th grade course for the Artificial Intelligence Pathway
AP Computer Science A	11.0160011 11.0160012	Y	11-12	Teacher Recommendation and a grade of 85 unweighted in AP Comp Sci Prin	AP Computer Science A is equivalent to a first-semester college computer science course, covering problem-solving, design strategies, data structures, algorithms, and the ethical and social aspects of computing. The course emphasizes object-oriented and imperative programming in Java, using methods that scale from simple to complex problems. Its curriculum aligns with many college-level CS1 courses. This is a 3rd year pathway course for the Computer Science Pathway and completes the Magnet Requirement
Introduction to Cybersecurity Embedded Course: Introduction to Hardware Technology	11.4810001 11.4810002 11.4480001 11.4480002	Y	10-11	Intro to Software Technology in the GT Block	Introduction to Cybersecurity is designed to provide students with the basic concepts and terminology of cybersecurity. The course examines how the concept of security integrates into the importance of user involvement, security training, ethics, trust, application of cybersecurity practices and devices, and best practices management. The fundamental skills cover internal and external threats to network security and design, how to enforce network level security policies, how to protect an organization's information, and a broad range of other topics. Introduction to Hardware Technology is the foundational course for Information Support & Services, Networking, and Cybersecurity pathways. This course is designed for high school students to understand, communicate, and adapt to a digital world as it impacts their personal lives, society, and the business world. Exposure to foundational knowledge in hardware, IT support, networks, and cybersecurity are all taught in a computer lab with hands-on activities and project-focused tasks. Students will not only understand the concepts but apply their knowledge to situations and defend their

					actions/decisions/choices through the knowledge and skills acquired in this course. This is a 2nd year pathway course for the Cybersecurity Pathway
Game Design: Animation and Simulation	11.4290001 11.4290002	Y	11-12	AP Computer Science Principles	Students in this course will learn the fundamental principles of game creation, exploring topics like game genres, play modes, player psychology, level design, storytelling, animation, mechanics, and game loops. Through hands-on, project-based challenges, students will create multiple games throughout the year to reinforce these concepts. This is a 3rd year pathway course for the Game Design Pathway and completes the Magnet Requirement
Embedding Computing	11.4270001 11.4270002	Y	11-12	AP Computer Science Principles	The demand for programming (software development) has gone well beyond desktop computers and the web, into a world of personal devices, smart cars, artificial intelligence, and more. These systems interact with us directly, as well as with each other. The things we take for granted are designed, built, and thought out. This course will focus on the interaction of programming and hardware, using data from various sensors in order to make decisions, find practical applications for programming, and design for real people to use. A common industry term to describe this work is the Internet of Things. Students will learn how the combination of programming and physical hardware accomplishes common but essential tasks throughout the world. This is a 3rd year pathway course for the Internet of Things Pathway and completes the Magnet Requirement
Advanced Cybersecurity	11.4820001 11.4820002	Y	11-12	Introduction to Cybersecurity	Advanced Cybersecurity is designed to provide students the advanced concepts and terminology of cybersecurity. The course explores the field of cybersecurity with updated content including new innovations in technology and methodologies. It builds on existing concepts introduced in Introduction to Cybersecurity and expands into malware threats, cryptography, organizational security, and wireless technologies. This is a 3rd year pathway course for the Cybersecurity Pathway and completes the Magnet Requirement
Programming, Games, Apps, and Society	11.4720001 11.4720002	Y	12	Completion of the following courses: AP CS A OR Game Design OR Embedded Computing AND	Are you ready to design and develop? The course is designed for high school students to strategize, design, and develop games and mobile and desktop applications that can be produced in the real world. Students will learn about life cycles of project development and use models to develop applications. Attention will be placed on how user interfaces affect the usability and effectiveness of a game or an application. Programming constructs will be employed which will allow students' applications to interact with "real world," stimuli. The course exposes students to privacy, legality, and security considerations with regards to the software industry.

				teacher recommendation AND Must be in grade 12	This is a 4th year IT senior elective course and can only be done after completing the Computer Science, Game Design, or Internet of Things Pathway

CTAE: Health Sciences

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics
Intro to Healthcare Science *Embedded in the GT Block	25.5210001 25.5210002	Y	9th	None	Introduction to Healthcare Science is the foundational course for all Health Science pathways and is a prerequisite for all other Healthcare Science pathway courses. This course will enable students to receive initial exposure to the many Healthcare Science careers as well as employability, communication, and technology skills necessary in the healthcare industry. This course will provide students with a competitive edge to be the better candidate for either entry into the healthcare global marketplace and/or the post-secondary institution of their choice to continue their education and training. *This is the first out of three pathway courses.* **Major Concepts and Topics Covered:** • Human growth and development • Interaction with patients and family members, • Health, wellness, and preventative care • Legal and ethical responsibilities of today's healthcare providers • Fundamental healthcare skills development • Microbiology • Basic life support and first aid
Essentials of Healthcare Embedded Course: Human Anatomy/Physiology	25.4400001 25.4400002 26.0730001 26.0730002	Y	10th	Introduction to Healthcare	Anatomy and Physiology is a vital part of most healthcare post-secondary education programs. **Major Concepts and Topics Covered:** • Anatomy and Physiology of each body system • Investigation of common diseases, disorders and emerging diseases. • Prevention of disease

					• Diagnosis and treatment • Medical terminology related to each system. This course provides an opportunity to demonstrate technical skills that enforce the goal of helping students make connections between medical procedures and the pathophysiology of diseases and disorders. *This is a second pathway course for students. They will take this if they plan to continue on to Surgical Technician, Emergency Medical Responder, Allied Health or Pharmacy Operations and Fundamentals.*
Therapeutic/Patient Care:					
~~Patient Care Fundamentals~~					This course is no longer offered at IA. Innovation Academy will not be offering Patient Care Fundamentals in the upcoming school year. As we continue to refine our healthcare pathway to reflect industry demand and maximize student preparation, we are recommending that students enroll in Allied Health instead. Allied Health provides broader, more flexible skill development, exposes students to a wider range of medical professions, and aligns more closely with IA's project-based, interdisciplinary approach to healthcare education. This course will better equip students for advanced pathway options, dual-enrollment opportunities, and competitive healthcare programs.
Emergency Medical Technician:					
Emergency Medical Responder	25.4500001 25.4500002	Y	11th or 12th	Introduction to Healthcare Science	The Emergency Medical Responder (EMR) course prepares the student to provide initial stabilizing care to the sick or injured prior to the arrival of Emergency Medical Services Professionals (EMS), and to assist EMS personnel in transporting patients for definitive care at an appropriate hospital/facility. **Major Concepts and Topics Covered:** • Medical Terminology • Anatomy & Physiology • Responder Safety • Incident Command • Blood-borne Pathogen Training • Basic Physical Assessment • Treatment of Trauma and Medical Emergencies • Cardiopulmonary Resuscitation

					• Automatic External Defibrillators (AEDs). • Additional Topics Include: Pathophysiology; Life Span Development; Public Health; Pharmacology; Airway Management; Respiration and Artificial Ventilation; Shock and Resuscitation; EMS Operations The course is a blend of lectures, hands on lab/learning, and practical scenario-based learning/testing. The course will include Healthcare Provider CPR/AED Certification from a Nationally Recognized Body (American Heart Association or Red Cross, etc.). *This is the third pathway course for the Emergency Medical Responder pathway.* **End of Pathway Assessment:** • This is an end of pathway course (3rd course out of 3) • The exam is The National Health Science Assessment
Therapeutic Services/Allied Health and Medicine					
Allied Health and Medicine	25.4370001 25.4370002	Y	11th or 12th	Intro to Healthcare And Essentials of Healthcare	This course is designed to offer seniors the opportunity to become effective and efficient multi-skilled healthcare providers as they develop a working knowledge of various allied health opportunities. Students focusing on a career path in the healthcare field may apply classroom/lab knowledge and skills according to their passion and interest in the health sciences field. The curriculum allows for student growth opportunities in area(s) of interest to the student. This course will result in completion of classroom simulated experience: this component of Allied Health is designed to give students practical application of previously studied knowledge and skills. Allied Health introduces students to a wide range of healthcare careers through hands-on skills, patient-centered practices, and real-world medical scenarios. This course builds essential clinical and professional foundations for all Health Science pathways at IA. Allied Health is recommended in lieu of Patient Care Fundamentals, which will not be offered next year. *This is the third pathway course in the Allied Health track.* **Major Concepts and Topics Covered:** • Medical Terminology • PPE • Lab Values

					• Vitals • Drug Schedules • Procedures • Phlebotomy • Pharmacology • Microbiology • Diseases • EKG Upon completion of this course and its prerequisites, this course meets the Certified Medical Assistant curriculum content as specified by the National Health career Association. Successful completion of the examination allows students to seek employment in the state of Georgia as a Certified Medical Assistant. **End of Pathway Assessment:** • This is an end of pathway course (3rd course out of 3) • The exam is The NHA CCMA Exam • Passing the exam (70% or higher) results in students being a certified medical assistant through the NHA
Therapeutic Services - Surgical Technician					
Surgical Technician I	25.4470001 25.4470002	Y	11th or 12th	Intro to Healthcare And Essentials to Healthcare	The goal of this course is to provide fundamental surgical technician skills including safety, infection control, pharmacology, surgical equipment, perioperative procedures, instruments, and sterilization. *This is the third pathway course in the Surgical Technician Pathway.* **Major Topics and Concepts Covered:** • Orientation to Surgical Technology, Legal Concepts, Risk Management, & Ethical Issues • The Surgical Patient, Physical Environment & Safety Standards • Biomedical Science • Preventing Perioperative Disease Transmission • Hemostasis, Emergency Situations, & All-Hazards Prep • Surgical Pharmacology & Anesthesia • Instrumentation, Equipment, & Supplies • Wound Healing, Sutures, Needles, & Stapling Devices • Surgical Case Management • Diagnostic Procedures

					• Multiple Surgical Specialties **End of Pathway Assessment:** • This is an end of pathway course (3rd course out of 3) • The exam is The National Health Science Assessment
Therapeutic Services - Pharmacy					
Pharmacy Operations and Fundamentals	25.45300001 25.45300002	Y	11th or 12th	Intro to Healthcare And Essentials to Healthcare	This course is an introduction to pharmacy technology professions, employment opportunities, and basic pre-pharmacy technician skills which may be utilized in either clinical or community settings such as retail, home health care, and ambulatory care pharmacies. This course is recommended for students planning on pursuing careers in the healthcare industry, which may require basic pharmaceutical knowledge, common healthcare mathematical applications, and/or technical proficiency in the administration of medications. An overview of prescription and nonprescription medications, classifications, actions, and interactions is provided while critical thinking skills are developed throughout the course from initial calculations/conversions of drug dosage forms to the simulation of regulating IV infusion rates. Technical skills in the preparation and administration of medications are practiced in simulated clinical labs. *This is the third pathway course in the Pharmacy Pathway.* **Major Topics and Concepts Covered:** • Ddifferent types of pharmacies, team members needed, and terminology used in a pharmaceutical setting. • Identification of natural and synthetic drugs. • History of government actions, law, practice, regulations, and standards. • Medical and Pharmaceutical terminology. • Identifying parts of a prescription, dosage forms, delivery systems, and routes of drug administration. • Formulate, calculate, and demonstrate proper drug doses and medication preparations for patient use (simulated). • Prescription processing, packaging, labeling, and dispensing (simulated) • IV set up and simulated simulation • Compounding techniques and measurements

					End of Pathway Assessment: • This is an end of pathway course (3rd course out of 3)
Biotechnology:					
Essentials of Biotechnology	25.5700001 25.5700002	Y	10	Introduction to Healthcare	This is the second course in the career pathway that introduces students to the broad understanding of the fundamentals of biotechnology and the impact on society. The knowledge and skills in this course provide a basic overview of current trends and careers in biotechnology, with an emphasis on basic laboratory skills, along with the business, regulatory, and ethical aspects of biotechnology. *This is the 2nd pathway course in the Biotechnology track.* **Major Concepts and Topics Covered:** • Safety, solutions, dilutions, micropipetting and spectrophotometry • DNA properties and protein synthesis. • Immune response • PCR, restriction enzymes, gel electrophoresis, CRISPR assay • Germ Theory
Applications of Biotechnology	25.5690001 25.5690002	Y	11	Introduction to Healthcare Science and Essentials of Biotechnology	This course further introduces students to the fundamentals of biotechnology. Included in this course are additional applications and techniques in biotechnology that expand and increase the student's comprehension of how biotechnology utilizes living systems to create products and enhance lives. In addition, laboratory applications learned in this course form the pivotal component distinguishing science theory from application in bioscience, like that of engineering and mathematics. Bioscience and the application of laboratory technique to the manipulation of living systems is a cornerstone of pharmaceutical, medical device, forensic science, environmental science, agriculture, alternative fuel, and green chemistry. *This is the third pathway course in the Biotechnology track*. **Major Concepts and Topics Covered:** • Protein Quantification and Enzyme Reactions • Tissue Culture • Drug Discovery • Immune Response Quantification • Synthetic Biology Gene Expression **End of Pathway Assessment:** • This is an end of pathway course (3rd course out of 3)

					• The exam is the
Biotechnology Internship/Independent Research	25.5750001 25.5750002	Y	12	Introduction to Healthcare Science, Essentials of Biotechnology, Applications of Biotechnology Grade of 85 or higher in pathway courses and successful completion of EOPA exam	This course (BIR) is designed for student biotechnology related research at Innovation Academy or off-site through an internship. Mentorship by the instructor will be given to students as they design and implement their own biology related research project. They will use tools and techniques developed from and related to the prerequisite biotechnology courses. Students will be expected to perform a literature review, develop an investigative question with a testable hypothesis, develop experimental methodology, quantitatively analyze and present data, draw conclusions from their data and present their data to an audience through a poster and a research paper. Students will have the option to work towards peer reviewed publishing depending on their research. Students who have off-site internships will develop their plan of research based on the internship site. They will be expected to document their research and present it through a poster and talk to an audience. BIR will be the culmination of biotechnology learned from the previous two years and is reliant on independent student research with the instructor as the mentor.
Dual Enrollment On Campus, Lab Off Campus: (see Dr. Dawes with all questions)					
BIOL 2113 - Anatomy and Physiology I (Lecture)	College Course Numbers	1 semester	12	Corequisite: BIOL 2113L (Lab) and ENGL 1101 (AP Lang Score of 3 or higher will satisfy this requirement)	• Introduces the anatomy and physiology of the human body. Emphasis is placed on the development of a systemic perspective of anatomical structures, physiological processes, and chemical principles related to physiological processes, and chemical principles related to physiology. Topics include body organization, cell structure and functions, tissue classifications, integumentary system, skeletal system, muscular system, and nervous and sensory systems. (associate degree-level course) • Students will take **BIOL 2113 (Lecture)** at IA Monday-Thursday. • **BIOL 2113L** Lab will be completed at Gwinnett Tech on a day of students' choosing (Evening, Friday, and Weekend options are available). Lab meets once per week.

					• **BIOL 2113L** will also satisfy IA Health Science Flex Friday requirements. • High School credits received = 1.5 • Gwinnett Tech credits received = 3 • *Please alot for 2 class periods in your schedule - one for lecture and one for lab
BIOL 2114 - Anatomy and Physiology II	College Course Numbers	1 semester	12	*Corequisite:* BIOL 2114L (Lab)	• Anatomy and Physiology II builds on the fundamentals studied in Anatomy and Physiology I by exploring the more intricate systems of the body. Topics include the endocrine system, cardiovascular system, blood and lymphatic system, immune system, respiratory system, digestive system, urinary system, and reproductive system. (associate degree-level course) • Students will take **BIOL 2114 (Lecture)** at IA Monday-Thursday. • **BIOL 2114L Lab** will be completed at Gwinnett Tech on a day of students' choosing (Evening, Friday, and Weekend options are available). Lab meets once per week. • **BIOL 2114L** will also satisfy IA Health Science Flex Friday requirements. • High School credits received = 1.5 • Gwinnett Tech credits received = 3 • *Please alot for 2 class periods in your schedule - one for lecture and one for lab

Research

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics
AP Seminar	23.0380001 23.0380002	Y	11	Lit/Comp 2 and Teacher recommendation	AP Seminar is a foundational, cross-curricular course exploring academic and real-world issues through diverse perspectives. Students analyze articles, research, and literary texts; synthesize

				(No summer or online credit recognized)	sources; write essays; and create presentations individually and in teams. Students must be prepared for college-level work, including frequent, timed writing. Regular attendance, engaged classroom participation, effective collaboration, and timely work submission are non-negotiable expectations of all students. Summer reading and writing are required for this course.
AP Research	23.0370001 23.0370002	Y	11 or 12	Scored 3 or above on the AP Seminar exam (if course taken) An unweighted 90 in AP Seminar and/or AP Science	AP Research, the second course in the AP Capstone experience, allows students to deeply explore an academic topic, problem, issue, or idea of individual interest. Students design, plan, and implement a yearlong investigation to address a research question. Through this inquiry, they further the skills they acquired in the AP Seminar course by learning research methodology, employing ethical research practices, and accessing, analyzing, and synthesizing information. Students reflect on their skill development, document their processes, and curate the artifacts of their scholarly work through a process and reflection portfolio. The course culminates in an academic paper of 4,000–5,000 words (accompanied by a performance, exhibit, or product where applicable) and a presentation with an oral defense. Course waivers are not available for AP Research.
IA Research Honors Pinnacle Project Program	40.0921041 40.0921042	Y	11	Lit/Comp II or AP Seminar	IA Research is an honors-level course in which students experience research through the lenses of both the scientific method and design thinking. Over the course of the year students will frame, execute and communicate their independent research through IA's Pinnacle Project using industry best practices. IA Research is analogous to AP Research in that research in any area - pathway, ELA, social sciences, science, math - is supported with the added flexibility College Board does not allow of partners or small groups and the opportunity to have community and expert mentors. There are specific subject area research sections such a science experimentation and fine arts. These cohorts are selected through the Pinnacle Prep program. Proposals are due the last week of February.

English Language Arts

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics
Literature and Composition I	23.0616001 23.0616002	Y	9	None	Literature and Composition I is a study of literary genres. Students will develop vocabulary and apply effective reading strategies to a wide variety of literary and informational texts; to learn characteristics of basic literary genres, including the novel, short story, poetry, drama, and nonfiction; to establish effective writing and research habits; and to refine language skills as they apply to writing, listening, speaking, and viewing. This course prepares students for college.
Literature and Composition I Honors	23.0616041 23.0616042	Y	9	Teacher recommendation and placement criteria	The honors level course has higher expectations and more rigorous coursework than the college preparatory level, including mandatory summer reading and written responses to literature. Literature and Composition I --Honors is a study of literary genres. Students will develop vocabulary and apply effective reading strategies to a wide variety of literary and informational texts; to learn characteristics of basic literary genres, including the novel, short story, poetry, drama, and nonfiction; to establish effective writing and research habits; and to refine language skills as they apply to writing, listening, speaking, and viewing. This course prepares students for college.
Literature and Composition II	23.0617001 23.0617002	Y	10	Literature and Composition I	Literature and Composition II is organized thematically. Students will develop vocabulary and apply effective reading strategies to a wide variety of literary and informational texts. They will learn about universal themes and symbols common to literary works, including the novel, short story, poetry, drama, and nonfiction. Students will build on effective writing and research habits as they refine language arts skills which they will apply to writing, listening, speaking, and viewing. This course prepares students for college.
Literature and Composition II Honors	23.0617041 23.0617042	Y	10	Literature and Composition 1 Honors or Teacher Recommendation	The course covers the same material as the on-level literature and composition II class, and covers other topics and skills specific to preparing students for the rigors of an Advanced Placement (AP) course will be integrated throughout the course.

AP Seminar	23.0380001 23.0380002	Y	10	Literature and Composition II and Teacher recommendation and Application Process No pre-requisite summer courses acknowledged	AP Seminar is a course that engages students in cross-curricular conversations that explore the complexities of academic and real-world topics and issues by analyzing divergent perspectives. Students learn to investigate a problem or issue, analyze arguments, compare different perspectives, synthesize information from multiple sources, and work alone and in a group to communicate their ideas.
11th American Literature and Composition	23.0510001 23.0510002	Y	11	Literature and Composition II	American Literature and Composition is a study of the major literary topics, themes, and movements in the history of the United States from pre-colonial times to present day. Students will focus on major literary forms of the emerging nation, analyze literary themes and trends, and both research and compose several papers, speeches, and presentations, using representative forms of discourse. This course prepares students for college.
11th American Literature and Composition Honors	23.3510041 23.3510042	Y	11	Literature and Composition II and Teacher Recommendation	The course covers the same material as the on-level American literature class, and covers other topics and skills specific to preparing students for the rigors of an Advanced Placement (AP) course will be integrated throughout the course.
AP Language & Composition	23.0530001 23.0530002	Y	11	AP Seminar or Teacher Recommendation	Advanced Placement English Language and Composition is an advanced college level study of the methods of development in expository, analytical, and argumentative writing. Students will write in a variety of composition methods, analyze rhetoric and writing styles, read and study American literature, and improve comprehension of rigorous texts in order to prepare for the Georgia End of Course exam in April and AP exam in May.
World Literature	23.0670001 23.0670002	Y	11 or 12	None	The course focuses on world literature and informational texts by and about people of diverse ethnic backgrounds. Students explore themes of linguistic and cultural diversity by comparing, contrasting, analyzing, and critiquing writing styles and universal themes. The students write argumentative, expository, narrative, analytical, and response essays. The research component is critical. The students observe and listen critically and respond appropriately to written and oral communication. Conventions are essential for reading, writing, and speaking. Instruction in language conventions will, therefore, occur within the context of reading, writing, and speaking rather than in isolation. The students

					understand and acquire new vocabulary and use it correctly in reading, writing, and speaking. This course reflects grade-level appropriate Georgia Standards of Excellence.
AP English Literature	23.0650001 23.0650002	Y	11 or 12	85 or above in previous English course and teacher recommendation	The AP English Literature and Composition course focuses on reading, analyzing, and writing about imaginative literature (fiction, poetry, drama) from various periods. Students engage in close reading and critical analysis of imaginative literature to deepen their understanding of the ways writers use language to provide both meaning and pleasure. As they read, students consider a work's structure, style, and themes, as well as its use of figurative language, imagery, and symbolism. Writing assignments include expository, analytical, and argumentative essays that require students to analyze and interpret literary works.
Dramatic Writing Embedded Course: Theatre Arts Lit I	52.0920001 52.0920002 52.0810011 52.0810012	Y	12	American Literature: 90 or above or American Literature Honors: 85 or above **AND** ELA teacher recommendation	Applies skills to culminate in creating and developing dramatic writing for theatrical media with special emphasis on film and television. Includes development of "writerly stance" by reading, viewing, and analyzing tests and visual media from a writer's point of view, with focus on understanding the construction process and including the application of conventions of standard English grammar and usage. **This honors-level course is recommended for motivated, engaged students with an interest in writing, researching, and collaborating on academic and creative works related to film, television, and drama.**

Mathematics

We highly encourage students NOT to take any math class over the summer to accelerate. Those students have not been prepared for the rigor for accelerated and/or AP math coursework. At IA, we encourage all students to take their math courses during the school year, IN PERSON.

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics
Algebra: Concepts and Connections	27.0811001 27.0811002	Y	9	Placement Criteria	This course is designed as the first course in a three-course series. Students will apply their algebraic and geometric reasoning skills to make sense of problems involving algebra, geometry, bivariate data, and statistics. This course focuses on algebraic, quantitative, geometric, graphical, and statistical reasoning. In this course, students will continue to enhance their algebraic reasoning skills when analyzing and applying a deep understanding of linear functions, sums and products of rational and irrational numbers, systems of linear inequalities, distance, midpoint, slope, area, perimeter, nonlinear equations and functions, quadratic expressions, equations and functions, exponential expressions, equations, and functions, and statistical reasoning.
Algebra: Concepts and Connections H	Coming Soon!	Y	9	8th Grade Math	Course description and curriculum alignment in development by Fulton County Schools in Spring 2026.
Geometry: Concepts and Connections	27.0821001 27.0821002	Y	9 and 10	Algebra: Concepts and Connections	This course is designed as the second course in a three-course series. This course enhances students' geometric, algebraic, graphical, and probabilistic reasoning skills. Students will apply their algebraic and geometric reasoning skills to make sense of problems involving geometry, trigonometry, algebra and probability. Students will continue to enhance their analytical geometry and reasoning skills when analyzing and applying a deep understanding of polynomial expressions, proofs, constructions, rigid motions and transformations, similarity, congruence, circles, right triangle trigonometry, geometric measurement, and conditional probability.
Geometry: Concepts and Connections H	27.0821041 27.0821042	Y	9 or 10	Algebra C&C or Algebra C&C H	Course description and curriculum alignment in development by Fulton County Schools in Spring 2026.
Advanced Algebra: Concepts and Connections	27.0831001 27.0831002	Y	10 or 11	Geometry: Concepts and Connections	Advanced Algebra: Concepts and Connections is the third course in a sequence of courses designed to ensure career and college readiness. It is intended to prepare students for fourth mathematics course options relevant to their post-secondary pursuits. High school course content

					standards are listed by big idea, including Data and Statistical Reasoning, Probabilistic Reasoning, Functional and Graphical Reasoning, Patterning and Algebraic Reasoning, and Geometric and Spatial Reasoning. In Advanced Algebra, students will continue to enhance their data and statistical reasoning skills as they learn specific ways to collect, critique, analyze and interpret data. Students will learn how to use matrices and linear programming to represent data and to solve contextually relevant problems. Students will strengthen their geometric and spatial reasoning skills as they learn how to solve trigonometric equations using the unit circle. In Advanced Algebra, students will further develop their functional and graphical reasoning as they explore and analyze structures and patterns for exponential, logarithmic, radical, polynomial, and rational expressions, equations and functions to further understand the world around them.
Advanced Algebra: Concepts and Connections H	27.0831041 27.0831042	Y	9, 10, or 11	Algebra C&C (H) and Geometry C&C (H)	Course description and curriculum alignment in development by Fulton County Schools in Spring 2026.
Enhanced Advanced Algebra and AP PreCalculus	27.0931001 27.0931002	Y	9, 10, 11 By teacher recommen-dation only	Geometry: Concepts and Connections	Enhanced Advanced Algebra & Precalculus is a thoughtful blend of Advanced Algebra: Concepts & Connections and Precalculus. Students will be provided the opportunity to develop a deep understanding of concepts in Algebra that are critical to the study of Calculus as well as an understanding of trigonometry and its applications. This course enhances students' geometric, algebraic, graphical, and probabilistic reasoning skills. Students will apply their algebraic and geometric reasoning skills to make sense of problems involving geometry, trigonometry, algebra, probability, and statistics. Students will continue to enhance their analytical geometry and reasoning skills when analyzing and applying a deep understanding of polynomial expressions, proofs, constructions, rigid motions and transformations, similarity, congruence, circles, right triangle trigonometry, geometric measurement, and conditional probability.
Advanced Mathematical Decision Making (AMDM)	27.0850001 27.0850002	Y	11 or 12	Advanced Algebra: Concepts and Connections (H)	Advanced Mathematical Decision Making (AMDM) is a high school mathematics course designed to prepare students for real-world scenarios in higher education and the workplace. It focuses on applying mathematical reasoning, statistics, and modeling to solve practical problems. Students explore topics such as personal finance, probability, data analysis, and network models to make informed decisions about issues they will face in their daily lives. This course moves beyond theoretical concepts to provide students with the quantitative reasoning and problem-solving skills necessary for navigating business, financial, and other complex situations.

AP PreCalculus	27.0741001 27.0741002	Y	10, 11, 12	Advanced Algebra: Concepts and Connections (H)	AP Precalculus covers topics designed to prepare students for Calculus and other higher level math courses. Topics include polynomial, rational, exponential, logarithmic, trigonometric, polar and parametric functions as well as limits, vectors, matrices and conic sections. Students planning to take AP Precalculus need to have essential skills including knowledge of functions (domain, range, excluded values), Algebraic skills (factoring, solving various types of equations), right triangle trigonometry, exponential and logarithmic functions, and critical thinking skills in math.
Pre-Calculus	27.0841001 27.0841002	Y	10, 11, 12	Advanced Algebra: Concepts and Connections	Precalculus is a fourth-year math option for students who have completed Advanced Algebra (or the equivalent). The course provides students with the opportunity to develop a deeper understanding of concepts in Algebra that are critical to the study of Calculus as well as an understanding of trigonometry and its applications. Throughout the course there should be a focus on notational fluency and the use of multiple representations. The course includes the study and analysis of piecewise and rational functions; limits and continuity as related to piecewise and rational functions; sequences and series with the incorporation of convergence and divergence; conic sections as implicitly defined curves; the six trigonometric functions and their inverses; applications of trigonometry such as modeling periodic phenomena, modeling with vectors and parametric equations, solving oblique triangles in contextual situations, graphing in the Polar Plane; solutions of trigonometric equations in a variety of contexts; and the manipulation and application of trigonometric identities.
Calculus	27.0780001 27.0780002	Y	11 or 12	PreCalculus or Enhanced Advanced Algebra and PreCalculus	Calculus is a fourth-year math option for students who have completed Pre-Calculus. The course provides students with the opportunity to develop an understanding of the derivative and its applications as well as the integral and its applications. Throughout the course there should be a focus on notational fluency and the use of multiple representations. The course includes the study and analysis of limits and continuity as applied to a variety of functions; the derivative as related to limits and continuity; various derivative rules such as product, quotient, and chain; applications of the derivative including curve analysis, applied max/min situations, related rate problems, and use of Mean Value Theorem; the definite integral as a limit of Riemann sums; properties of definite integrals; the Fundamental Theorem of Calculus as it relates derivatives and integrals; techniques of integration including u-substitution; and applications of the integral including solving separable differential equations, finding a particular solution curve given an initial condition, area between curves

					on a coordinate plane, and average value situations. Topics should be analyzed in multiple ways, to include verbal and written, numerical, algebraic, and graphical presentations. Instruction and assessment should include the appropriate use of technology. Concepts should be introduced and investigated, where appropriate, in the context of realistic phenomena.
AP Calculus AB	27.0720001 27.0720002	Y	11-12	PreCalculus or Enhanced Advanced Algebra and PreCalculus	AP Calculus AB is a year-long course comparable to the first semester of calculus courses offered at colleges or universities. Students need to have a thorough knowledge of college preparatory mathematics including, trigonometry, analytic geometry, equations and graphs, and lines and conics. Students taking this course should be adequately prepared to study elementary functions; limits and continuity; and differential and integral calculus. Calculus AB is concerned with developing students' understanding of the concepts of calculus and providing experience with its methods and applications. The course emphasizes a multi-representational approach to calculus, with concepts, problems, and results being expressed graphically, numerically, analytically, and verbally. You'll perform experiments and investigations and solve problems by applying your knowledge and skills.
AP Calculus BC	27.0730001 27.0730002	Y	11-12	PreCalculus or Enhanced Advanced Algebra and PreCalculus	AP Calculus BC is a year-long course comparable to the first two semesters of calculus courses offered at colleges or universities. Students need to have a thorough knowledge of college preparatory mathematics including, trigonometry, analytic geometry, equations and graphs, and lines and conics. Students taking this course should be adequately prepared to study elementary functions; limits and continuity; differential and integral calculus; calculus of parametric, polar, and vector functions; and series and polynomial approximations. Calculus BC is concerned with developing students' understanding of the concepts of calculus and providing experience with its methods and applications. The course emphasizes a multi-representational approach to calculus, with concepts, problems, and results being expressed graphically, numerically, analytically, and verbally. You'll perform experiments and investigations and solve problems by applying your knowledge and skills.
AP Statistics	27.0740001 27.0740002	Y	10, 11, 12	Advanced Algebra: Concepts and Connections or Enhanced Advanced	Learn about the major concepts and tools used for collecting, analyzing, and drawing conclusions from data. You'll explore statistics through discussion and activities, and you'll design surveys and experiments. Skills you will learn in this course include selecting methods for collecting or analyzing data, describing patterns, trends, associations, and relationships in data, using probability and simulation to describe probability

				Algebra and PreCalculus	distributions and define uncertainty in statistical inference, and using statistical reasoning to draw appropriate conclusions and justify claims.

Science

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics
Biology	26.0120001 26.0120002	Y	9	None	Students will identify patterns, processes, and relationships of living organisms including the interdependence of organisms, the relationship of matter, energy, and organization in living systems, the behavior of organisms, and biological evolution. Students will investigate biological concepts through experiences in laboratories and field work using the process of inquiry.
Biology Honors	26.0120041 26.0120042	Y	9	Placement Criteria	Honors Biology is a lab science course designed for the advanced academic student who has the ability to critically analyze and apply biological concepts, the ability to gain information on the scientific processes through critical reading, and one who possesses strong organizational skills. This challenging course focuses on cytology, ecology, genetics, evolution, taxonomy, microbiology, botany, and zoology. Honors Biology students are required to complete an in-depth, individual literature review and research paper during the first semester. This is a lab inclusive course with 25% of the time expected to be spent in the lab.
AP Biology	26.0140001 26.0140002	Y	10-12	Biology Honors	In this Advanced Placement course, students will further develop an understanding of biology through inquiry-based investigations exploring the topics of evolution, cellular processes—energy and communication, genetics, information transfer, ecology, and interactions.
Earth Systems	40.0640001 40.0640002	Y	10-12	None	Students investigate connections among Earth's systems (atmosphere, hydrosphere, and geosphere); the Earth's landscapes, ecology, and resources; phenomena fundamental to geology and physical geography (including the early history of Earth, plate tectonics, landform evolution, the Earth's geologic record, weather and climate, and history of life on Earth).
Honors Chemistry	40.0510041 40.0510042	Y	10-12	Biology Honors	This course is a lab science course. Skills needed for this course are the ability to apply past learning to new concepts; the demonstration of abstract and higher-level thinking; the ability to perform algebraic manipulations easily; the ability to read critically; self-motivation; and experience in writing formal lab reports. Topics covered in this yearlong

					course are characteristics of science, the nature and classification of matter, stoichiometry, and conservation of matter, the atom, the atomic theory, the periodic table, and reaction rates. Honors Chemistry students are required to complete numerous labs with analysis and a project during each semester. This is a lab-inclusive course with 25% of the time expected to be spent in the lab.
AP Chemistry	40.0530001 40.0530002	Y	11-12	Honors Chemistry	In this Advanced Placement course, students will investigate the structure of matter, bonding and intermolecular forces, chemical reactions, kinetics, and thermodynamics and chemical equilibrium through the application of science practices and laboratory investigations.
Physics	40.0810001 40.0810002	Y	10-12	Algebra, Geometry and Trigonometry	This lab science course provides students with the necessary knowledge and skills in physics. Physics extends the physical sciences to more abstract concepts including interactions of matter and energy, velocity, acceleration, forces, energy, momentum, thermodynamics, charge, electricity, magnetism, waves, light, optics, and subatomic physics. The subject is treated both conceptually and mathematically. Concepts are investigated through laboratory experiences and fieldwork designed for students to develop appropriate knowledge and skills in science as inquiry. This is a lab-inclusive course with 25% of the time expected to be spent in the lab.
Honors Physics	Coming Soon!	Y	10-12	Algebra, Geometry and Trigonometry Concurrently enrolled in pre-calculus	Explores concepts in greater depth and moves at a faster pace than regular Physics. Covers topics beyond the standard curriculum, such as more advanced mechanics or aspects of quantum & Nuclear physics. Higher level of complexity, speed, and analytical depth to prepare students for AP Physics.
AP Physics I	40.0831001 40.0831002	Y	10-12	Pre-calculus (90 unweighted) Advanced Algebra (90 unweighted)	AP Physics 1 is an Algebra-based Advanced Placement course (College Level Course) that introduces college-level physics units which explores Kinematics, Dynamics with Newtonian Mechanics (rotational dynamics and angular momentum), Conservation of Energy (including work, energy, and power) and Momentum. This college level course uses conceptual understanding and applications of physics in the real world to understand the mechanisms of physics.
AP Physics C Mechanics	40.0841001 40.0841002	.5	11-12	AP Physics 1	Mechanics is a one-semester, calculus-based, college-level physics course, especially appropriate for students planning to specialize in or major in one of the physical sciences or engineering. Students cultivate their understanding of physics through classroom study and activities as well as hands-on laboratory work as they explore concepts like change, force interactions, fields, and conservation.

AP Physics C Electricity	40.0842001 40.0842002	.5	11-12	AP Physics 1	AP Physics C: Electricity and Magnetism is a one-semester, calculus-based, college-level physics course, especially appropriate for students planning to specialize in or major in one of the physical sciences or engineering. Students cultivate their understanding of physics through classroom study and activities as well as hands-on laboratory work as they explore concepts like change, force interactions, fields, and conservation.
AP Environmental Science	26.0620001 26.0620002	Y	10-12	Honors Biology	In this Advanced Placement course, students investigate ecosystems, human population, major global problems, energy resources, pollution, sustaining biodiversity and ecological integrity, and the environment as it relates to society. This course integrates previous knowledge from biology and chemistry.
Forensic Science	43.4520001 43.4520002	Y	11-12	None	This course provides students with an opportunity to explore the basic processes and principles of forensic science as it relates to criminal investigations. Students will learn the importance of identification, collection, and processing of evidence during a criminal investigation. Students will learn the roles, legal responsibilities, and challenges of forensic and criminal investigators. Included in this course is the importance of preserving and documenting a crime scene and enabling an investigator to analyze evidence and its relationship to the crime. Students will also study interviews and interrogations and how statements are used as evidence in court.
Organic Chemistry Honors	40.0570041 40.0570042	Y	11 or 12	Honors Chemistry	In this course, students learn to recognize and name organic functional groups. The course focuses on the properties and reactions of common classes of organic compounds; the relationship between the structures of organic compounds and their physical and chemical properties. Students learn about systems to represent organic molecules, stereochemistry, how structure affects physical properties, drawing resonance forms with proper arrow convention, organic acid-base reactions, substitution and elimination reactions and one-step syntheses. In the laboratory, students investigate how structure affects physical properties such as reactivity, boiling point, melting point, optical rotation, and solubility. Students also learn how to perform fundamental techniques such as crystallization, filtration, distillation, refractive index, extraction, thin-layer, column, and gas chromatography.
Astronomy	40.0570001 40.0570002	Y	11 or 12	None	This course will provide the student with an introduction to the concepts of modern astronomy, the origin and history of the Universe, and the formation of the Earth and the solar system. Students will compare the Earth's properties with those of the other planets and explore how the heavens have influenced human thought and action. The course gives a

					description of astronomical phenomena using the laws of physics. The course treats many standard topics including planets, stars, the Milky Way and other galaxies, and black holes. Laboratory exercises include experiments in light properties, measurement of radiation from celestial sources, and observations at local observatories and/or planetariums. Authentic Research opportunities will be provided.
Super Biology	1 credit of Honors Biology-Fall Semester, 1 credit of AP Biology-Spring Semester	Y	9	A two-tiered invitation only course based on 8th grade accelerated math and accelerated science unweighted grades, Students should also be concurrently enrolled for pre-calculus as their 9th grade math course	Full year course, 90-minute block with additional meetings on Fridays, Biology End of Course Test and Advanced placement biology exam will be given. Invited students will have to take a readiness assessment placement test in order to move forward with placement for the course
Super Chemistry	1 credit of Honors Chemistry-Fall Semester, 1 credit of AP Chemistry-Spring Semester	Y	10	Teacher recommendation only	Full year course, 90-minute block with additional meetings on Fridays, Advanced placement chemistry exam in May
Super Physics	AP Physics 1-Fall Semester, AP Physics 2-Spring Semester	Y	11	Teacher recommendation only, Students should also be concurrently enrolled or completed Calculus	Full year course, 90-minute block with additional meetings on Fridays, Advanced placement chemistry exam in May. Recommended students will have to take a readiness assessment placement test to move forward with placement for the course

Social Studies

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics
American Government/Civics	45.0570001	.5	9	None	An in-depth study of the American political system. This course focuses on the foundation, principles and structure of the American system of government, examines the role of political parties, social factors as they relate to the role of the citizen, and analyzes the decision-making process that are a part of the system of American political behavior. This course meets the state's Citizenship requirement for graduation.
Economics	45.0610001	.5	9	None	An introductory course into the principles of economics. The course includes topics related to Fundamental Economic Concepts, Microeconomics Concepts, Macroeconomics Concepts, International Economics, and Personal Finance Economics.
AP Human Geography	45.0520001 45.0520002	Y	9-12 elective	None	Systematic study of patterns and processes that have shaped human understanding, use, and alteration of the Earth's surface. This course is equivalent to an introductory college-level course in human geography. It introduces students to the systematic study of patterns and processes that have shaped human understanding, use, and alteration of Earth's surface. Students employ spatial concepts and landscape analysis to examine socioeconomic organization and its environmental consequences, and the methods and tools geographers use in their research and application.
World History	45.0830001 45.0830002	Y	10	None	Prehistoric culture, ancient civilizations, classical civilizations, the medieval world, the Age of Exploration, Enlightenment, French Revolution, decline of colonial empires in America, Industrial Revolution, nationalism and imperialism, totalitarianism, WWI, WWII, and the modern world.
AP World History	45.0811001 45.0811002	Y	10	None	The evolution of global processes and contacts in inter- action with different types of human societies; the nature of changes in international frame- works and their causes and consequences, as well as comparisons among major societies
U.S. History	45.0810001 45.0810002	Y	11	None	Colonization, the revolutionary and colonial eras, manifest destiny, Civil War and reconstruction, urbanization and Industrialism, progressive era, imperialism, WWI & WWII, The Cold War, Vietnam, and the Decades of 1950 – 2000.
AP U.S. History	45.0811001 45.0811002	Y	11	None	Multicultural heritage, Colonial period, American Revolution, Jacksonian Democracy and sectionalism, Civil War and Reconstruction, Triumph of the American Nation, Gilded Age, Progressivism and immigration, Great Depression and New Deal, Labor movement, Civil Rights and women's movement, World Wars I and II, Cold War, and New World Order.
AP Government and Politics	45.0570001 45.0570002	Y	9 or 12	None	Political philosophies that influenced the foundations of U.S. government and why countries develop different forms of government globally; U.S. constitutional principles and the branches of government;

					and factors influencing the political process. Students will construct and evaluate arguments, use documents and other primary source data to analyze point of view and understand and interpret information, and write document-based and comparative analysis essays
AP Psychology	45.0160001 45.0160002	Y	11 or 12	None	Explore the ideas, theories, and methods of the scientific study of behavior and mental processes. You'll examine the concepts of psychology through reading and discussion and you'll analyze data from psychological research studies.
AP Microeconomics *Paired with AP Macroeconomics	45.0630001	.5	11 or 12		AP Microeconomics is an introductory college-level microeconomics course. Students cultivate their understanding of the principles that apply to the functions of individual economic decision-makers by using principles and models to describe economic situations and predict and explain outcomes with graphs, charts, and data as they explore concepts like scarcity and markets; costs, benefits, and marginal analysis; production choices and behavior; and market inefficiency and public policy.
AP Macroeconomics *Paired with AP Microeconomics	45.0620001	.5	11 or 12		Explore the principles of economics that apply to an economic system as a whole. You'll use graphs, charts, and data to analyze, describe, and explain economic concepts.
AP Art History		1	11 or 12	None	AP Art History offers a mix of art and history that's more than just memorizing dates and names. Spanning cultures across all continents and from prehistoric to contemporary times, in this course you'll explore the different social and cultural beliefs, political and economic systems, and scientific and technological advancements that have led to the creation of well-known sculptures, paintings, buildings, and many other types of art.
US History in Film (Must be taken as a one semester course paired with AP Govt Comp)	45.0812001	.5	11 or 12	None	Movies throughout the age of film have sought to entertain and teach. This course will investigate and evaluate films about American History based on their accuracy and their role as historical evidence. A major goal of this course is determining what is valid in contemporary films and historical dramas and what these films say about the people who create them, the politics behind their creation, and how they reflect the values, ideas, and larger historical issues of the times in which they were created. Students in this course will 1) view movies on various topics, 2) participate in discussion about the values and limitations of film in conveying history, and 3) write essays comparing film evidence to information in more traditional sources, such as articles, textbooks, and critical commentaries, 4) and compare multiple films on the same topic

					to examine different points of view. Some of the films viewed in this course will be rated "R". Parental permission is required to take this course.
AP Govt/Politics Comp (Must be taken as a one semester course paired with US History in Film)	45.0530001	.5	11 or 12	None	The AP Comparative Government and Politics course detailed in this framework reflects what comparative political science teachers, professors, and researchers agree that a college-level comparative government and politics course should teach students to do: define and describe major political concepts, analyze patterns of political processes and behavior and their consequences, and compare and contrast political institutions and processes across countries. The AP Comparative Government and Politics Course and Exam Description defines what representative colleges and universities typically expect students to know and be able to do in order to earn college credit or placement. Students practice the skills used by comparative political 17 | P a g e scientists by studying data, political writings from different countries, and the processes and outcomes of politics in a variety of country settings. Students will show mastery of these skills on the AP Exam through various means, including applying concepts, analyzing data, comparing countries, and writing political science arguments. Students study six countries in AP Comparative Government and Politics: China, Iran, Mexico, Nigeria, Russia, and the United Kingdom. Students interested in government, politics, international relations, and/or in working for nongovernmental organizations internationally will enjoy this course.

TAG

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics
Directed Study	70.2320000	Y	10-12	Approval of TAG teacher	Directed Studies may be taken in all academic areas. Student and teacher will write a curriculum contract that lists goals, objectives, and requirements.
Gifted Internship I, II, III, IV	70.221000, 70.2220000, 70.2230000, 70.2240000	Y	Preference give to 12th graders 30 slots each for fall and spring	Approval for course will be based upon the TAG student's academic, attendance, and	TAG students are assigned to work with professionals in a field that they are considering as a career. They have the opportunity to gain experience and insight into the business world and make decisions about career goals. Students will leave the school for one or two periods a day. The Internship will count as either one or two of their regular courses during the semester.

				discipline records; flexibility in schedule; teacher recommendation; and application.	

Physical Education

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics
FVS Health	17.3110001	**.5**	9-12	None	This course explores the mental, physical and social aspects of life and how each contributes to total health and well-being. Emphasizes safety, nutrition, mental health, substance abuse prevention, disease prevention, environmental health, family life education, health careers, consumer health , and community health.
FVS Personal Fitness	36.3510001	**.5**	9-12	None	Introduces instruction in methods to attain a healthy level of physical fitness; implements a lifetime fitness program based on a personal fitness assessment and stresses strength, muscular endurance, flexibility, body composition, and cardiovascular endurance; includes instruction in fitness principles, nutrition, fad diets, weight control, stress management, adherence strategies, and consumer information; and promotes self-awareness and responsibility for fitness.

Electives

Course Title	Course #	Term	Grade(s)	Prerequisite(s)	Major Topics
Beginning Music Technology	53.0221001 53.0221002	Y	9-12	None	The goal of beginning music technology is to give students the tools, framework, and context to create musical projects and recordings using modern music production practices. Areas of concentration will include work recording techniques on Digital Audio Workstations, MIDI controllers, keyboard and piano technique, composition, songwriting, music for visual media, sound design with virtual and analog instruments, synthesizer manipulation, song analysis, audio editing, mixing, transcription, musical literacy, and music theory (harmony and melody).

Intermediate Music Technology	53.0222001 53.0222002	Y	9 – 12	Beginning Music Technology	Continue to develop skills introduced in Beginning Music Technology. Topics will include standard recording techniques, synthesizer/MIDI techniques, plugins and advanced effects, music theory, artist development, and social media management. This course will largely be project-based, and will culminate in large-scale recording projects (mixtapes, EPs, albums, etc.).
Yearbook	23.0320003 23.0320004	Y	9-12	By application only	Yearbook is a full-year elective course that gives students marketable experience in print media publishing. In this course students will gain skills in graphic design, copywriting, editing, and photography while developing communication skills through reporting, writing, class discussions, presentations, and publications. Advised by the teacher, students will create the yearbook as they learn how photography and the written word combine to inform society. Students will gain useful, real-world skills in time management, marketing, teamwork, communication, and design principles.
Audio Visual Film & Technology I	10.5181001 10.5181002	Y	9-12	None	This course is designed to prepare students to become content creators. Students will learn skills related to digital story telling, composition, camera equipment, script writing, production teams, thumbnail graphic design, lighting, recording and editing video and audio. Once complete students will have the skills to create and publish videos and podcasts.
Journalism I	23.032000	Y	9 – 12	Audio Visual Film and Technology I	Students in this class are part of IATV, a weekly journalistic video produced by students for students. Students will be guided in the process of interviewing people of interest to them and how best to tell their story. Through out the year students will participate in film festivals, podcasts and digital media contests. Students will advance their skills in audio and lighting production from AVTF 1. Once complete students will be able to produce a video on a deadline and at a higher level.
Journalism II	23.0330001	Y	9 – 12	By application only	Students in this course take a leadership role within planning IAtv content and special video assignments. Students will have the opportunity to travel to Student Television Network conference and compete with over 3,000 video students in contests. Example competitions: Silent Film, News Package, Music Video, Short Film etc. Additionally, students will have the opportunity to work on a capstone content creation project related to a passion of theirs.
Work-Based Learning	Course numbers	Y	12	Meet WBL requirements	Work-based Learning (WBL) is designed to provide relevant experiences in all career areas that will assist students in making

	depend on area of study				decisions concerning their future. Work-Based Learning placements represent the pinnacle of the Career-Related Education experience. Enrollment in a work-based learning course is an extension of the student's work in their College and Career pathway. Credit earned for enrollment in work-based learning may count toward graduation as part of the student's cluster or pathway. To qualify for a WBL placement, a student must be in grades 11 or 12 and at least 16 years old. Students must also have a defined Career Pathway in order to participate in the Work-Based component of Career-Related Education. This is especially important for successful completion of a student's pathway in that their job placement is directly related to the curriculum of the pathway classes they have completed or in which they are concurrently enrolled.
Peer Facilitation	35.0410001 35.0410002	Y	11 and 12	Contract signed with assigned staff member	Students provide assistance to selected teachers in their classrooms and must be able to interact with students and adults in a variety of settings.
Beacon Writing Staff	23.0320007 23.0320008	Y	10-12	By application only	Students work to design, plan, and write articles for the FCS IA Beacon. In this course students will gain skills in graphic design, copywriting, editing, and photography while developing communication skills through reporting, writing, class discussions, presentations, and publications. They will also develop skills in interviewing and communicating with their features and audience.
Intro to Art (Must be taken with Design Fundamentals)	50.0211001	One semester only	10-12	None	This course is the pre-requisite for all other studio art courses. Introduction to Art is an entry-level class that establishes a standard and consistent foundation in the discipline of visual art. Students will be introduced to all aspects of visual art including, but not limited to, art as personal communication, drawing, sculpture, ceramics, design, aesthetics, careers, art criticism and art history. Students develop basic skills that increase critical thinking, problem solving, self-evaluation and the ability to complete long-term projects.
Design Fundamentals (Must be taken with Intro to Art)	50.0212001	One semester only	10-12	Intro to Art	Enhances level-one skills in art history, art criticism, aesthetic judgment, and studio production. Emphasizes and reinforces knowledge and application of the design elements and their relationship to the principles of design. Explores different two- and three-dimensional art media and processes. Investigates master artworks to increase awareness and to examine the role of art and the artist in past and contemporary societies.
Draw Paint I	50.0313001	One semester only	10-12	Intro to Art	**Drawing & Painting I** will instruct students in fundamental drawing skills and prepare them to make the transition to painting. Approaches include contour, value to model form, gesture,

(Must be taken with Draw Paint II)					perspective, and color. Students work with drawing media such as pencil, charcoal, conte and oil pastels. Art history, criticism and aesthetics are incorporated with studio production of drawings and paintings. In addition to learning a life-long skill, drawing courses help increase observation skills, self-discipline, ability to evaluate one's own performance, problem solving abilities, and ability to complete long-term projects.
Draw Paint II (Must be taken with Draw Paint I)	50.0314001	One semester only	10-12	Intro to Art and Draw Paint I	**Drawing & Painting II** develops fundamental painting skills and continues to strengthen composition and drawing skills. This course enhances level one skills in technique and provides further exploration of drawing media. Drawing skills and critical analysis skills are reinforced for responding to master drawings of different historical styles and periods. This course addresses increasingly complex drawing and painting problems and the development of personal style. Art history, criticism, and aesthetics are incorporated with studio production of drawings and paintings.

Appendix 1: Course Progression

ENGLISH PATHWAY



SCIENCE PATHWAYS



**Note: Even if a student has the Physical Science credit from middle school, as a STEM magnet school, we still recommend a Physics class in High School.*

***Note: Even if a student has taken on level Biology in middle school, we recommend Biology H in 9th grade to ensure the correct level of high school science rigor.*

MATH PATHWAYS

Middle School	9th Grade	10th Grade	11th Grade	12th Grade
8th Grade Math	Algebra C&C Algebra C&C H	Geometry C&C Geometry C&C H	Advanced Algebra C&C Advanced Algebra C&C H	Pre-Calculus AP Pre-Calculus AP Statistics

Middle School	9th Grade	10th Grade	11th Grade	12th Grade
Algebra C&C H	Geometry C&C Geometry C&C H	Advanced Algebra C&C Advanced Algebra C&C H	Pre-Calculus AP Pre-Calculus AP Statistics	Calculus AP Calculus AB AP Calculus BC AP Statistics AMDM

Middle School	9th Grade	10th Grade	11th Grade	12th Grade
Algebra C&C H	Geometry C&C Geometry C&C H	Enhanced Advanced Algebra and AP Pre-Calculus	Calculus AP Calculus AB AP Calculus BC AP Statistics AMDM	Dual Enrollment Math Any math course not yet taken.

By teacher recommendation only

Middle School	9th Grade	10th Grade	11th Grade	12th Grade
Algebra C&C H Geometry C&C H	Advanced Algebra C&C Advanced Algebra C&C H	Pre-Calculus AP Pre-Calculus AP Statistics	Calculus AP Calculus AB AP Calculus BC AP Statistics AMDM	Dual Enrollment Math Any math course not yet taken.

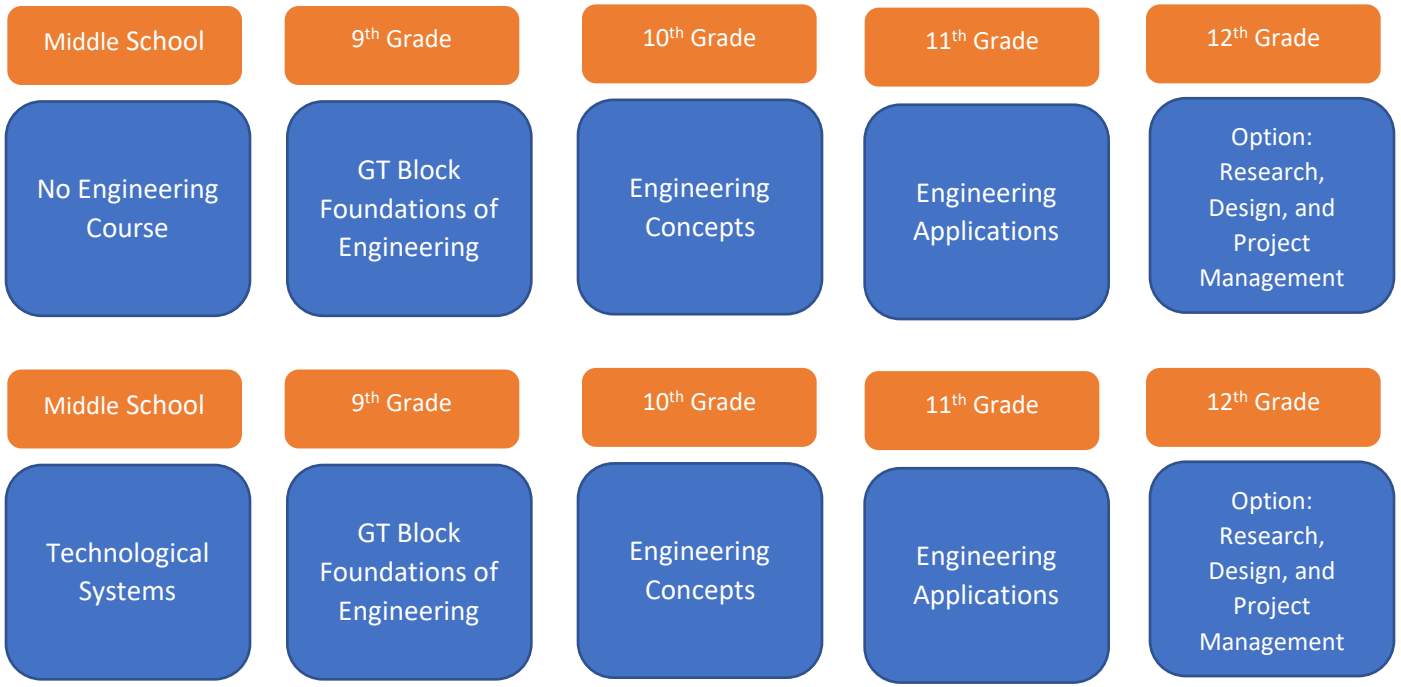
Middle School	9th Grade	10th Grade	11th Grade	12th Grade
Algebra C&C H Geometry C&C H Advanced Algebra C&C H	Pre-Calculus AP Pre-Calculus AP Statistics	Calculus AP Calculus AB AP Calculus BC AP Statistics AMDM	Dual Enrollment Math Any math course not yet taken.	Dual Enrollment Math Any math course not yet taken.

SOCIAL STUDIES PATHWAYS

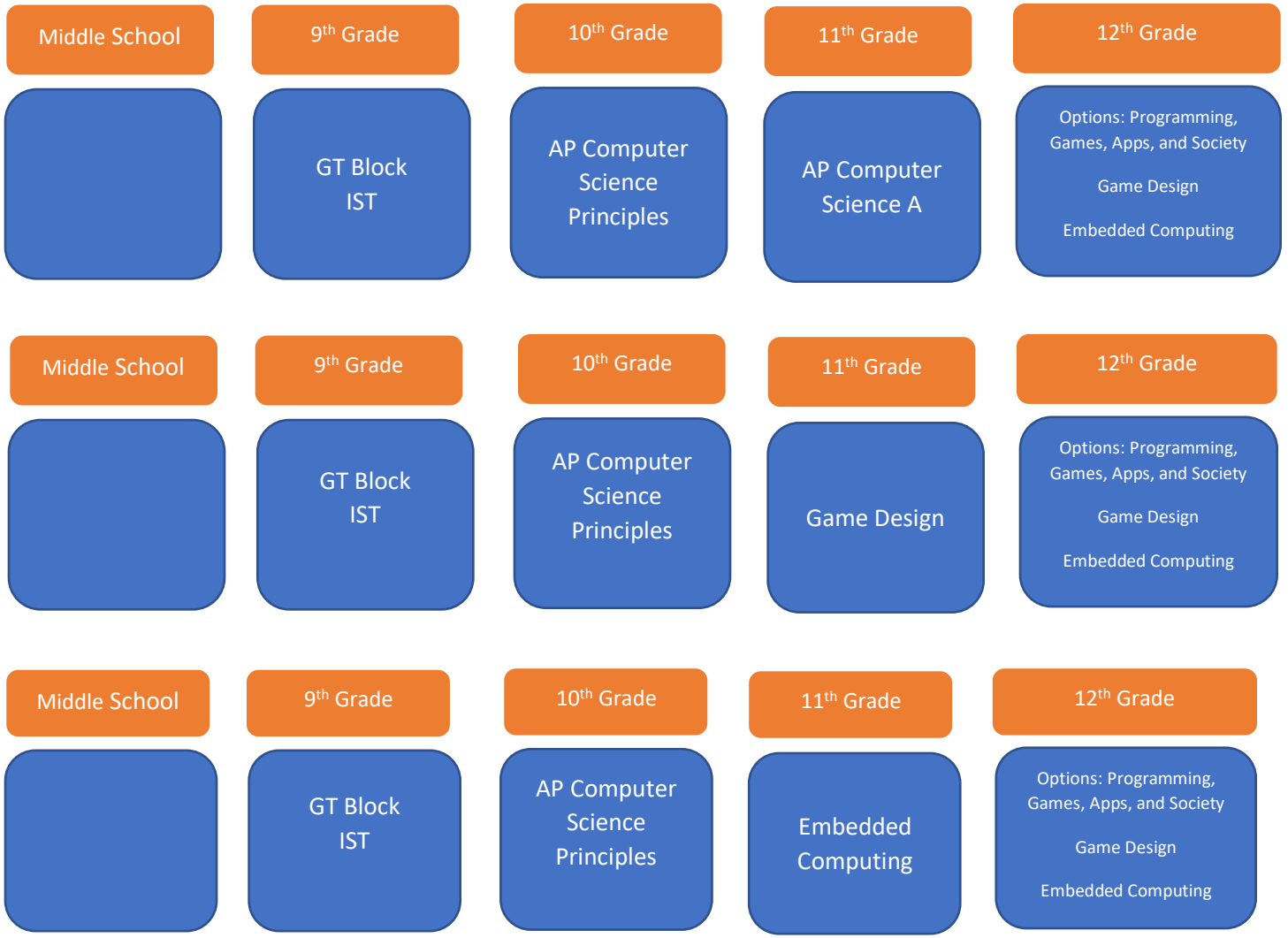


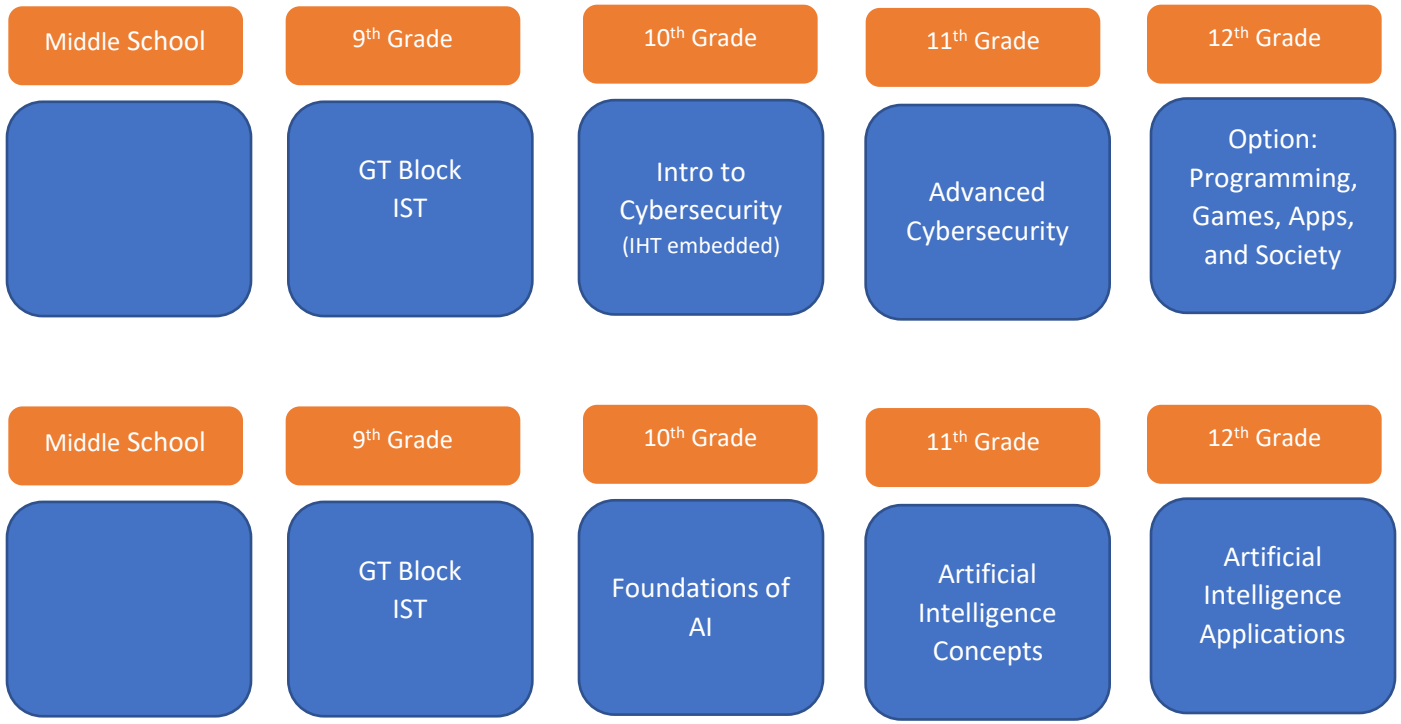
Social Studies Elective: AP Art History

Engineering Pathway

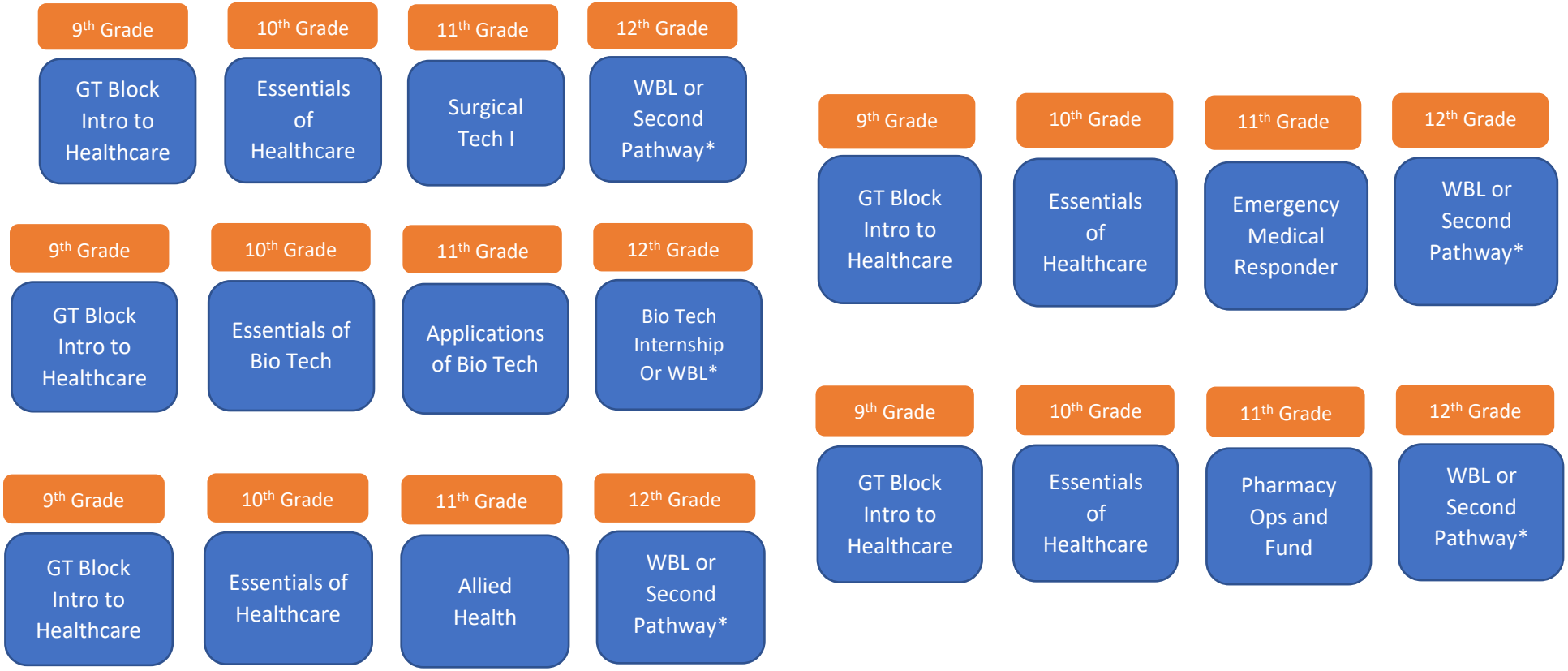


IT Pathways


Middle School
9th Grade
10th Grade
11th Grade
12th Grade
GT Block IST
AP Computer Science Principles
AP Computer Science A
Options: Programming, Games, Apps, and Society
Game Design
Embedded Computing
Middle School
9th Grade
10th Grade
11th Grade
12th Grade
GT Block IST
AP Computer Science Principles
Game Design
Options: Programming, Games, Apps, and Society
Game Design
Embedded Computing
Middle School
9th Grade
10th Grade
11th Grade
12th Grade
GT Block IST
AP Computer Science Principles
Embedded Computing
Options: Programming, Games, Apps, and Society
Game Design
Embedded Computing


Middle School
9th Grade
10th Grade
11th Grade
12th Grade
GT Block IST
Intro to Cybersecurity (IHT embedded)
Advanced Cybersecurity
Option: Programming, Games, Apps, and Society
Middle School
9th Grade
10th Grade
11th Grade
12th Grade
GT Block IST
Foundations of AI
Artificial Intelligence Concepts
Artificial Intelligence Applications

Healthcare Pathways



***Students do not have to take a Healthcare course senior year. It is optional.**

Appendix 2: Graduation Status Report

Graduation Status Report

23 Credits Required for Graduation

Name:

Total Credits Earned to Date:

Year of Graduation:

English **4 Credits**		**Math** **4 Credits**		**Science** **4 Credits**		**Social Studies** **3 Credits**		**Health/Personal Fitness** **1 Credit**				**Electives** **4 Credits minimum**
☐	Lit/Comp I – A	☐		☐	Biology / Honors	☐	American Government	☐	General Health	☐	☐	Intro HCS
☐	Lit/Comp I – B	☐		☐	Biology / Honors			☐ *	Personal Fitness Waiver? ☐ See Electives	☐	☐	Intro Software Tech or IDT
☐	Lit/Comp II – A	☐		☐	Physical Science or Physics A	☐	World History Sem. 1	**World Language, Fine Arts, or Career Tech 3 Credits**		☐	☐	Fundamentals of Engineering
☐	Lit/Comp II – B	☐		☐	Physical Science or Physics B	☐	World History Sem. 2	☐		☐	☐	Pathway Year 2
☐	11th Lit – A AP Lang / Other	☐		☐	Earth/Envs/Chem	☐	US History Sem. 1	☐		☐	☐	Pathway Year 3
☐	11th Lit – B AP Lang / Other	☐		☐	Earth/Envs/Chem	☐	US History Sem. 2	☐		☐	☐	Extra Choices
☐	AP Lit / ML / DW / Adv Comp	☐		☐		☐	Economics	☐		☐	☐	Extra Choices
☐	AP Lit / ML / DW / Adv Comp	☐		☐				☐		☐	☐	Pinnacle Project – IA requirement
								☐				